MOTOROLA

MC6850
1.0 MHz
MC68A50
1.5 MHz
MC68B50
2.0 MHz

ASYNCHRONOUS COMMUNICATIONS INTERFACE ADAPTER (ACIA)

The MC6850 Asynchronous Communications Interface Adapter provides the data formatting and control to interface serial asynchronous data communications information to bus organized systems such as the MC6800 Microprocessing Unit.

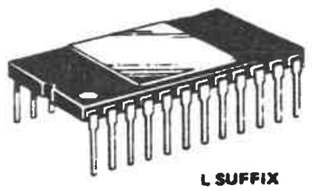
The bus interface of the MC6850 includes select, enable, read/write, interrupt and bus interface logic to allow data transfer over an 8-bit bi-directional data bus. The parallel data of the bus system is serially transmitted and received by the asynchronous data interface, with proper formatting and error checking. The functional configuration of the ACIA is programmed via the data bus during system initialization. A programmable Control Register provides variable word lengths, clock division ratios, transmit control, receive control, and interrupt control. For peripheral or modem operation three control lines are provided. These lines allow the ACIA to interface directly with the MC6860L 0-600 bps digital modem.

- Eight and Nine-Bit Transmission
- Optional Even and Odd Parity
- Parity, Overrun and Framing Error Checking
- Programmable Control Register
- Optional ÷1, ÷16, and ÷64 Clock Modes
- Up to 500 kbps Transmission
- False Start Bit Deletion
- Peripheral/Modem Control Functions
- Double Buffered
- One or Two Stop Bit Operation

MOS

(N-CHANNEL, SILICON-GATE)

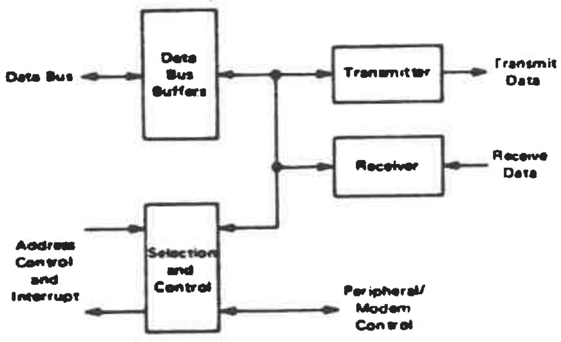
ASYNCHRONOUS COMMUNICATIONS INTERFACE ADAPTER



L SUFFIX
CERAMIC PACKAGE
CASE 716

NOT SHOWN: P SUFFIX
PLASTIC PACKAGE
CASE 709

MC6850 ASYNCHRONOUS COMMUNICATIONS INTERFACE ADAPTER BLOCK DIAGRAM



ORDERING INFORMATION

Speed	Device	Temperature Range
1.0 MHz	MC6850P,L	0 to +70°C
	MC6850CP,CL	−40 to +85°C
MIL-STD-883B MIL-STD-883C	MC6850BJCS MC6850CJCS	−55 to +125°C
1.5 MHz	MC68A50P,L	0 to +70°C
	MC68A50CP,CL	−40 to +85°C
2.0 MHz	MC68B50P,L	*0 to +70°C

 DS 9493 R1

MAXIMUM RATINGS

Rating		Symbol	Value	Unit
Supply Voltage		V_{CC}	-0.3 to +7.0	Vdc
Input Voltage		V_{in}	-0.3 to +7.0	Vdc
Operating Temperature Range		T_A	0 to +70	°C
Storage Temperature Range		T_{stg}	-55 to +150	°C
Thermal Resistance	Plastic	θ_{JA}	120	°C/W
	Ceramic		60	

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

ELECTRICAL CHARACTERISTICS (V_{CC} = 5.0 V ±5%, V_{SS} = 0, T_A = 0 to 70°C unless otherwise noted.)

Characteristic		Symbol	Min	Typ	Max	Unit
Input High Voltage		V_{IH}	V_{SS} + 2.0	–	V_{CC}	Vdc
Input Low Voltage		V_{IL}	V_{SS} -0.3	–	V_{SS} + 0.8	Vdc
Input Leakage Current (V_{in} = 0 to 5.25 Vdc)	R/W, CS0, CS1, $\overline{CS2}$, Enable, RS, Rx D, Rx C, $\overline{CTS}$, $\overline{DCD}$	I_{in}	–	1.0	2.5	µAdc
Three-State (Off State) Input Current (V_{in} = 0.4 to 2.4 Vdc)	D0-D7	I_{TSI}	–	2.0	10	µAdc
Output High Voltage	D0-D7	V_{OH}				Vdc
(I_{Load} = -205 µAdc, Enable Pulse Width ≤25 µs)			V_{SS} + 2.4	–	–	
(I_{Load} = -100 µAdc, Enable Pulse Width ≤25 µs)	Tx Data, $\overline{RTS}$		V_{SS} + 2.4	–	–	
Output Low Voltage (I_{Load} = 1.6 mAdc, Enable Pulse Width ≤25 µs)		V_{OL}	–	–	V_{SS} + 0.4	Vdc
Output Leakage Current (Off State) (V_{OH} = 2.4 Vdc)	$\overline{IRQ}$	I_{LOH}	–	1.0	10	µAdc
Power Dissipation		P_D	–	300	525	mW
Input Capacitance		C_{in}				pF
(V_{in} = 0, T_A = 25°C, f = 1.0 MHz)	D0-D7		–	10	12.5	
	E, Tx Clk, Rx Clk, R/W, RS, Rx Data, CS0, CS1, $\overline{CS2}$, $\overline{CTS}$, $\overline{DCD}$		–	7.0	7.5	
Output Capacitance (V_{in} = 0, T_A = 25°C, f = 1.0 MHz)	$\overline{RTS}$, Tx Data	C_{out}	–	–	10	pF
	$\overline{IRQ}$		–	–	5.0	
Minimum Clock Pulse Width, Low (Figure 1)	÷16, ÷64 Modes	PW_{CL}	600	–	–	ns
Minimum Clock Pulse Width, High (Figure 2)	÷16, ÷64 Modes	PW_{CH}	600	–	–	ns
Clock Frequency	÷1 Mode	f_C	–	–	500	kHz
	÷16, ÷64 Modes		–	–	800	
Clock-to-Data Delay for Transmitter (Figure 3)		t_{TDD}	–	–	1.0	µs
Receive Data Setup Time (Figure 4)	÷1 Mode	t_{RDSU}	500	–	–	ns
Receive Data Hold Time (Figure 5)	÷1 Mode	t_{RDH}	500	–	–	ns
$\overline{\text{Interrupt Request}}$ Release Time (Figure 6)		t_{IR}	–	–	1 2	µs
$\overline{\text{Request-to-Send}}$ Delay Time (Figure 6)		t_{RTS}	–	–	1.0	µs
Input Transition Times (Except Enable)		t_r, t_f	–	–	1.0*	µs

*1.0 µs or 10% of the pulse width, whichever is smaller.

BUS TIMING CHARACTERISTICS

Characteristic	Symbol	MC6850 Min	MC6850 Max	MC68A50 Min	MC68A50 Max	MC68B50 Min	MC68B50 Max	Unit
READ (Figures 7 and 9)								
Enable Cycle Time	t_{cycE}	1.0	–	0.666	–	0.500	–	µs
Enable Pulse Width, High	PW_{EH}	0.45	25	0.28	25	0.22	25	µs
Enable Pulse Width, Low	PW_{EL}	0.43	–	0.28	–	0.21	–	µs
Setup Time, Address and R/W valid to Enable positive transistion	t_{AS}	160	–	140	–	70	–	ns
Data Delay Time	t_{DDR}	–	320	–	220	–	180	ns
Data Hold Time	t_H	10	–	10	–	10	–	ns
Address Hold Time	t_{AH}	10	–	10	–	10	–	ns
Rise and Fall Time for Enable input	t_{Er}, t_{Ef}	–	25	–	25	–	25	ns
WRITE (Figures 8 and 9)								
Enable Cycle Time	t_{cycE}	1.0	–	0.666	–	500	–	µs
Enable Pulse Width, High	PW_{EH}	0.45	25	0.28	25	0.22	25	µs
Enable Pulse Width, Low	PW_{EL}	0.43	–	0.28	–	0.21	–	µs
Setup Time, Address and R/W valid to Enable positive transition	t_{AS}	160	–	140	–	70	–	ns
Data Setup Time	t_{DSW}	195	–	80	–	60	–	ns
Data Hold Time	t_H	10	–	10	–	10	–	ns
Address Hold Time	t_{AH}	10	–	10	–	10	–	ns
Rise and Fall Time for Enable input	t_{Er}, t_{Ef}	–	25	–	25	–	25	ns

MOTOROLA

MC6808

Advance Information

MICROPROCESSOR WITH CLOCK

The MC6808 is a monolithic 8-bit microprocessor that contains all the registers and accumulators of the present MC6800 plus an internal clock oscillator and driver on the same chip.

The MC6808 is completely software-compatible with the MC6800 as well as the entire M6800 family of parts. Hence the MC6808 is expandable to 65K words.

This very cost-effective MPU allows the designer to use the MC6808 in consumer as well as industrial applications without sacrificing industrial specifications.

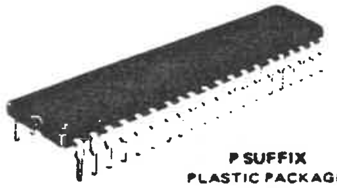
- On-Chip Clock Circuit
- Software-Compatible with the MC6800
- Expandable to 65K words
- Standard TTL-Compatible Inputs and Outputs
- 8-Bit Word Size
- 16-Bit Memory Addressing
- Interrupt Capability

MOS

(N-CHANNEL, SILICON-GATE, DEPLETION LOAD)

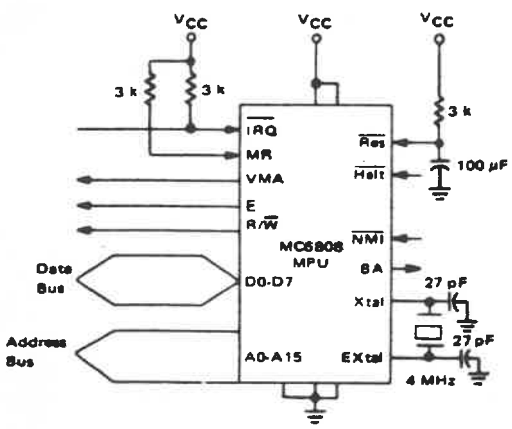
MICROPROCESSOR WITH CLOCK

L SUFFIX
CERAMIC PACKAGE
CASE 715



P SUFFIX
PLASTIC PACKAGE
CASE 711

FIGURE 1 – TYPICAL MICROPROCESSOR INTERFACE



PIN ASSIGNMENT

1	V_{SS}	Reset	40
2	Halt	EXtal	39
3	MR	Xtal	38
4	IRQ	E	37
5	VMA	V_{SS}	36
6	NMI	V_{CC}	35
7	BA	R/W	34
8	V_{CC}	D0	33
9	A0	D1	32
10	A1	D2	31
11	A2	D3	30
12	A3	D4	29
13	A4	D5	28
14	A5	D6	27
15	A6	D7	26
16	A7	A15	25
17	A8	A14	24
18	A9	A13	23
19	A10	A12	22
20	A11	V_{SS}	21

This is advance information and specifications are subject to change without notice.

ADI-806